6G-Oriented LDPC-Coded Faster-Than-Nyquist Signaling: Code Design and Performance Analysis

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***Abstract*—This paper focuses on the design and performance analysis of faster-than-Nyquist (FTN) signaling employing enhanced 5G low-density parity-check (LDPC) codes, oriented toward the requirements of future 6G systems. We propose the extrinsic information transfer (EXIT) chart analysis for the LDPC-coded FTN system based on the Ungerboeck observation model, where the input-output mutual information function of the detector is approximated using least squares fitting. With the proposed EXIT chart analysis, we explore the thresholds and decoding performance of different LDPC codes (regular codes, irregular codes and protograph codes) in both Nyquist and FTN systems, revealing two important observational findings for FTN signaling: 1) unlike Nyquist systems, where certain 5G New Radio (NR)-like information puncturing can enhance the decoding threshold and performance, we observe that in the FTN setting considered in this paper such puncturing leads to performance degradation; 2) unlike Nyquist systems, the parity-check matrix of LDPC codes optimized for FTN signaling tends to be relatively sparser within comparable ensembles, due to the intentionally introduced inter-symbol interference (ISI). Based on these findings, we develop tailored LDPC codes for FTN signaling by applying the masking operation to the base matrix of the standard 5G LDPC codes, aiming to achieve a lower decoding threshold and thereby better decoding performance. Moreover, the raptor-like structure and rate compatibility are preserved in the proposed LDPC codes, and the encoder and decoder are reused with only minor modifications. Numerical results show that: 1) all simulation results align with the decoding thresholds obtained by the proposed EXIT chart analysis, confirming the effectiveness of the analysis; 2) for the FTN system, the tailored LDPC codes outperform standard 5G LDPC codes, achieving over 0.4 dB coding gain and approaching (slightly exceeding) the constrained Nyquist capacity; 3) under the same spectral efficiency, FTN with tailored LDPC codes performs better than standard 5G LDPC codes with Nyquist signaling, demonstrating a coding gain of up to 0.6 dB; 4) proposed LDPC codes with the FTN signaling achieve better performance compared to existing high-performance codes specifically designed for FTN signaling.**

***Index Terms*—Extrinsic information transfer (EXIT) chart, faster-than-Nyquist (FTN), low-density parity-check (LDPC) codes.**

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I. Introduction

With the rapid deployment of the fifth-generation (5G) networks [2], global research efforts have shifted toward the sixth-generation (6G) technologies to address rapidly growing demands [3]. A core challenge for the deployment of the 6G network is the stringent requirements on the data rate that is vital in supporting new applications, such as holographic communications [4] and immersive extended reality [5]. Unfortunately, the conventional Nyquist signaling, while inherently avoiding inter-symbol interference (ISI) through strict orthogonality among transmitted symbols, is insufficient in meeting such stringent requirements. In contrast, faster-than-Nyquist (FTN) signaling can enhance the spectral efficiency (SE) by intentionally introducing controlled ISI via adjusting the symbol rate [6], making it a prominent and promising waveform for next-generation communication.

FTN signaling was first conceptualized by Shannon in 1948 [7] and formally introduced by Mazo in 1975 [6]. The basic idea of FTN signaling is to transmit information symbols at rates faster than the Nyquist rate, therefore introducing an additional degree of freedom in the design and optimization of coded-modulation systems—potentially achieving higher SE than systems restricted by the Nyquist criterion. In fact, FTN signaling has a Shannon capacity close to the unconstrained capacity of the shaping pulse and has been shown to outperform the Shannon capacity of Nyquist signaling with non-sinc shaping pulses [8]. As a result, FTN signaling has garnered significant attention for various communication applications, including satellite communications [9], optical communications [10], and wireless communications [11], [12], [13].

Due to the inherent ISI of the FTN system, conventional channel codes designed for Nyquist systems may not perform well under FTN transmissions. Therefore, designing tailored channel codes is crucial for enhancing the performance of the FTN system. In [14], rate-1/2 convolutional codes optimized for FTN signaling have been developed through convergence analysis and exhaustive search. Furthermore, a special type of convolutional codes named output-retainable convolutional codes has been proposed in [15], characterized by a code trellis intentionally overlapped with the ISI trellis to enable low-complexity joint detection and decoding schemes. In [16], block Markov superposition transmission (BMST) codes [17], [18] are applied to FTN systems to enable low-complexity windowed decoding.

In addition to the convolutional codes and coupled codes, low-density parity-check (LDPC) codes, known for their good decoding performance, have become the primary coding scheme in modern communication systems and are widely adopted in modern communication standards like 5G New Radio (NR) [19], DVB-S2 [20] and IEEE 802.16e (WiMAX) [21]. Considering the significance and broad application of LDPC codes, initial studies have also investigated to design of LDPC codes for FTN systems. Specifically, inspired by the progressive edge growth (PEG) algorithm, a unified searching approach was introduced in [22] to construct quasi-cyclic (QC)-LDPC codes with larger girth, enhancing their performance in FTN systems. In [23], a uniform shortening scheme for LDPC codes is introduced to mitigate ISI effects and thereby improve the performance of FTN systems. The 5G LDPC codes have been directly employed in FTN systems to obtain simulation performance [24].

However, the aforementioned LDPC code constructions lack analytical support, particularly regarding decoding threshold analysis. Consequently, the underlying theory of LDPC code optimization for FTN signaling is still yet to be developed. The decoding threshold is an effective metric for code optimization, which can be obtained via extrinsic information transfer (EXIT) chart analysis [25], [26], [27], [28], [29]. In [30], a numerical upper bound on the achievable information rate of LDPC-coded FTN systems was proposed. However, it cannot provide an accurate decoding threshold due to the complexity of the FTN detector, as there is no direct or approximate formula to characterize the input-output mutual information (MI) for the FTN detector. To the best of our knowledge, a complete EXIT chart analysis for LDPC-coded FTN systems has not yet been conducted. Consequently, decoding threshold results remain unavailable for the LDPC-coded FTN systems, making it rather challenging to completely optimize the LDPC code design tailored for FTN systems.

To address the aforementioned limitations, this paper first develops a complete EXIT chart analysis for LDPC-coded FTN systems to predict decoding thresholds, which can be leveraged for optimizing LDPC codes for FTN signaling. Based on the developed EXIT chart analysis, we then focus on the analysis and design of tailored LDPC codes for FTN signaling without compromising their rate-compatible characteristics.

The main contributions of this paper are listed as follows:

1) We employ the least-squares fitting method to approximate the input-output MI function of the FTN detector. Based on this approximation, we develop the complete EXIT chart analysis for the LDPC-coded FTN system to predict the decoding thresholds.
2) By adopting the proposed EXIT chart analysis, we compare the thresholds and decoding performance for LDPC codes with different structures (regular, irregular, and protograph codes) in both FTN and Nyquist systems. The comparative results demonstrate a clear consistency between the decoding thresholds and the actual decoding performance, which validates the correctness of the proposed analysis. Furthermore, these results reveal an important fact that LDPC codes performing well in Nyquist systems do not necessarily perform well in FTN systems. Particularly, we find two key observational findings related to 5G NR-like information puncturing, and appropriate sparsity of the parity-check matrix, which are important for designing high-performance LDPC codes for FTN systems.
3) Based on the above observational findings, we propose tailored LDPC codes by applying the masking operation [31] to the base matrix of the standard 5G LDPC codes, thereby reducing the average node degree. This modification results in a lower decoding threshold and hence also better decoding performance compared to standard 5G LDPC codes for FTN signaling. Particularly, the key structural features of the standard 5G LDPC code, such as the raptor-like structure and rate compatibility, are preserved in the proposed tailored LDPC codes so that the encoder and decoder only require minor modifications from the 5G standard. Numerical results demonstrate that: i) for the FTN system, the proposed LDPC codes outperform the standard 5G LDPC codes, which is consistent with our EXIT chart analysis, resulting in over 0.4 dB coding gains; ii) under the same spectral efficiency constraint, FTN with tailored LDPC codes outperforms the standard 5G LDPC codes with Nyquist signaling by up to 0.6 dB; iii) the proposed LDPC codes with the FTN signaling achieve better error performance compared to other codes specifically optimized for FTN signaling.

The rest of this paper is organized as follows: In Sec. II, the 5G LDPC codes and the system model of the LDPC-coded FTN system are presented. Sec. III provides the EXIT chart analysis of the LDPC-coded FTN system, comparing different LDPC codes for both Nyquist and FTN systems. The proposed LDPC codes for FTN systems are discussed in Sec. III. In Sec. IV, numerical results for both Nyquist and FTN systems are presented. The conclusion is given in Sec. V.

II. Preliminaries

A. 5G LDPC Codes

The 5G LDPC codes are a type of rate-compatible quasi-cyclic (QC) LDPC codes with raptor-like structure. There are

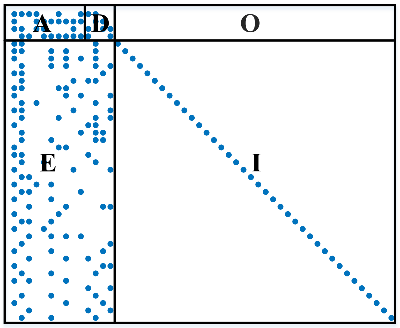

Fig. 1. The scatter diagram of BG2 for 5G LDPC codes, where parts $\mathbf{A}$ and $\mathbf{D}$ correspond to the kernel part of the 5G LDPC codes and the other parts are employed for generating the extended parity bits.

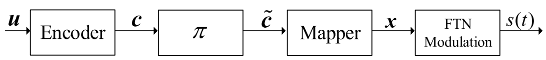

Fig. 2. The diagram of the considered coded FTN transmitter.

two base matrices corresponding to two base graphs, i.e., base graph 1 (BG1) and base graph 2 (BG2), which are specifically designed for different information lengths and code rates [19]. The scatter diagram of BG2 with size of $(m_b+2)\times(n_b+2)$ is shown in Fig. 1, where each dot represents a non-zero point, and each of columns corresponds to one variable-node type, since in the base graph each variable node has a specific degree and adjacency pattern. Given the base matrices, the parity-check matrices $\mathbf{H}$ of size $(m_b+2)Z\times(n_b+2)Z$ can be constructed by the lifting operator with a lifting factor Z. As a result, each dot in the scatter diagram is transformed into a permutation matrix of size $Z\times Z$, and other positions are all zero. Typically, to further improve the decoding performance for Nyquist signaling, the first $2Z$ bits (corresponding to the first two columns of the base graph) of the 5G LDPC codes are punctured.

B. LDPC-Coded Faster-Than-Nyquist Signaling

The diagram of the LDPC-coded FTN transmitter is shown in Fig. 2. At the transmitter, the information sequence $\boldsymbol{u}\in\mathbb{F}_2^k$ is first encoded by the encoder of an LDPC code $\mathcal{C}[n,k]$, resulting in a codeword $\boldsymbol{c}\in\mathbb{F}_2^n$. Then, the codeword $\boldsymbol{c}$ is interleaved by a random interleaver $\boldsymbol{\pi}$ yielding $\widetilde{\boldsymbol{c}}$, which is rewritten as $\widetilde{\boldsymbol{c}}=(\widetilde{\boldsymbol{c}}_0,\widetilde{\boldsymbol{c}}_1,\cdots,\widetilde{\boldsymbol{c}}_{N-1})$ with $\widetilde{\boldsymbol{c}}_t\in\mathbb{F}_2^m$ and $N=n/m$ (for simplicity, assuming that n/m is integer). The interleaved codeword $\widetilde{\boldsymbol{c}}$ is mapped to the symbol vector $\boldsymbol{x}$ with $x_t=\phi(\widetilde{\boldsymbol{c}}_t)$ and $\boldsymbol{x}$ is then modulated by the FTN modulator, resulting in the FTN signal as

$$s(t)=\sum_{l=0}^{N-1}x_l h(t-l\tau T),\tag{1}$$

where x_l denotes the l-th energy-normalized symbol, $h(t)$ is the shaping pulse, T is the Nyquist symbol period, and $\tau\in(0,1]$ is the FTN compression factor [6]. Without loss of generality, in this paper, we apply the T-orthogonal root raised cosine (RRC) pulse with a roll-off factor $\beta=0.3$ as the shaping pulse $h(t)$.

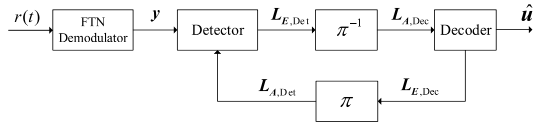

Fig. 3. The diagram of the considered coded FTN receiver.

Assume that the FTN signal $s(t)$ is transmitted over an additive white Gaussian noise (AWGN) channel, yielding the received signal $r(t)=s(t)+n(t)$, where $n(t)$ is the AWGN process with one-sided power spectral density (PSD) N_0. Then, the discrete received sequence $\boldsymbol{y}$ is obtained by first matched-filtering with respect to $h(t)$ and then sampling at the FTN rate [32], which leads to

$$\boldsymbol{y}=\mathbf{G}\boldsymbol{x}+\boldsymbol{\eta},\tag{2}$$

where $\mathbf{G}$ is a Toeplitz matrix characterizing the ISI among transmitted symbols [33], i.e.,

$$\mathbf{G}=\begin{pmatrix} g_0 & g_{-1} & \cdots & g_{-L} & 0 & 0 & 0 & \cdots \\ g_1 & g_0 & g_{-1} & \cdots & g_{-L} & 0 & 0 & \cdots \\ \vdots & g_1 & g_0 & g_{-1} & \cdots & g_{-L} & 0 & \cdots \\ g_L & \vdots & g_1 & g_0 & g_{-1} & \cdots & g_{-L} & \cdots \\ 0 & g_L & \vdots & g_1 & g_0 & g_{-1} & \cdots & \cdots \\ \vdots & 0 & g_L & \vdots & g_1 & g_0 & g_{-1} & \cdots \\ \vdots & \vdots & 0 & g_L & \vdots & g_1 & g_0 & \cdots \end{pmatrix}.\tag{3}$$

The elements $\{g_i\}$ in (3) are the ISI coefficients satisfying

$$g_i=\int_{-\infty}^{\infty}h(t)h^*(t-i\tau T)\mathrm{d}t,\quad -L\leq i\leq L.\tag{4}$$

The term L in (4) denotes the ISI truncation length, under the assumption that only the $2L+1$ central taps around g_0 with have significant energy, while the remaining taps are negligible and thus omitted for simplicity. The $\boldsymbol{\eta}$ in (2) is the colored noise vector with zero-mean and the covariance matrix $\mathbb{E}[\boldsymbol{\eta}\boldsymbol{\eta}^{\mathrm{H}}]=N_0\mathbf{G}$, where $(\cdot)^{\mathrm{H}}$ denotes the Hermitian transpose. We set $L=5$ to strike a balance between detection accuracy and computational complexity, and we use this setting consistently throughout this paper.

The diagram of the Turbo-equalization based LDPC-coded FTN receiver is shown in Fig. 3. The estimated information sequence $\hat{\boldsymbol{u}}$ is obtained by performing outer iterations (between detector and decoder) up to $J_{\max}$ between the detector and decoder, and inner iterations (for LDPC decoding) up to $I_{\max}$, where a parity-check termination is utilized for the early termination of the iterative decoder. In the paper, we apply the sum-product algorithm (SPA) based on the Ungerboeck observation model for the detection [34]. This method has shown promising decoding performance for detecting FTN signals with relatively low computational complexity [35].

Formally, denote by $\boldsymbol{L}_{A,\mathrm{Det}}$ the input *a priori* log-likelihood-ratios (LLRs) to the detector and by $\boldsymbol{L}_{E,\mathrm{Det}}$ the output extrinsic LLRs from the detector. Similarly, we use $\boldsymbol{L}_{A,\mathrm{Dec}}$ and $\boldsymbol{L}_{E,\mathrm{Dec}}$ to denote the *a priori* LLRs and extrinsic LLRs of the decoder, respectively. The FTN detector is implemented based on the SPA over a factor graph. At the block level, the detector takes as input (i) the received signal block $\boldsymbol{y}$ and (ii) the *a priori* LLRs $\boldsymbol{L}_{A,\text{Det}}$ provided by the LDPC decoder. It outputs the extrinsic LLRs $\boldsymbol{L}_{E,\text{Det}}$, which are passed to the LDPC decoder for further decoding. Specifically, the SPA detector constructs a factor graph where variable nodes represent transmitted symbols and factor nodes correspond to conditional probability distributions based on the received signal. Messages are iteratively exchanged between these nodes, approximating the posterior probability of each symbol. The extrinsic LLR for the i-th bit of symbol $\tilde{\boldsymbol{c}}_t$ is computed as

$$L_{E,\text{Det}}(t,i) = \ln \frac{\sum_{\tilde{\boldsymbol{c}}_t \in \mathcal{L}_i^0} p(\boldsymbol{y}|x=\phi(\tilde{\boldsymbol{c}}_t))p(\tilde{\boldsymbol{c}}_t|\boldsymbol{L}_{A,\text{Det}})}{\sum_{\tilde{\boldsymbol{c}}_t \in \mathcal{L}_i^1} p(\boldsymbol{y}|x=\phi(\tilde{\boldsymbol{c}}_t))p(\tilde{\boldsymbol{c}}_t|\boldsymbol{L}_{A,\text{Det}})} - L_{A,\text{Det}}(t,i), \quad (5)$$

where $\mathcal{L}_i^b = \left\{\tilde{\boldsymbol{c}}_t \in \mathbb{F}_2^m \mid \tilde{c}_t^{(i)} = b\right\}$ for $i = 0,1,\cdots,m-1$. Given the LLR sequence $\boldsymbol{L}_{A,\text{Dec}}$, the extrinsic information of the j-th bit of a codeword from the decoder can be calculated as,

$$L_{E,\text{Dec}}(j) = \ln \frac{p(c_j=0|\boldsymbol{L}_{A,\text{Dec}})}{p(c_j=1|\boldsymbol{L}_{A,\text{Dec}})} - L_{A,\text{Dec}}(j), \quad (6)$$

where $j = 0,1,\cdots,n-1$.

In the iterations, the extrinsic messages are passed between the FTN detector and the LDPC decoder as in the bit-interleaved coded modulation with iterative decoding (BICM-ID) system [36], [37]. For a simple comparison, the Nyquist signaling system adopts the same iterative detection and decoding setting as in the FTN case.

C. Fair Comparison Between FTN and Nyquist Signaling

For infinite-length comparisons, a criterion often used in theoretical analyses (e.g., [8]) is to match the PSD of the FTN and Nyquist signals, where the signal becomes wide-sense stationary. The PSD of the FTN signal is given by

$$S_x(f) = \frac{\mathbb{E}[|x|^2]}{\tau T}|H(f)|^2, \quad (7)$$

where $H(f)$ is the Fourier transform of the shaping pulse $h(t)$. The term $\mathbb{E}[|x|^2]/(\tau T)$ represents the average transmit power. Clearly, for fixed symbol energy $\mathbb{E}[|x|^2]$, the FTN signal has higher power than Nyquist signaling (where $\tau = 1$) due to its shorter symbol interval. This implies that, under the same PSD normalization, FTN must use less energy per symbol than Nyquist signaling.

For practical finite-length comparisons, fixing the total energy per codeword is a more appropriate and fair criterion. Thus, both FTN and Nyquist signals are modulated by sequences of symbols with roughly the same length, and the average symbol energy for both transmissions is normalized to the same in this paper. Such an arrangement can make sure that the total transmitted energy is the same for both FTN and Nyquist signals. However, the effective duration of the FTN signal is shorter due to the higher symbol rate, yielding a greater spectral efficiency.

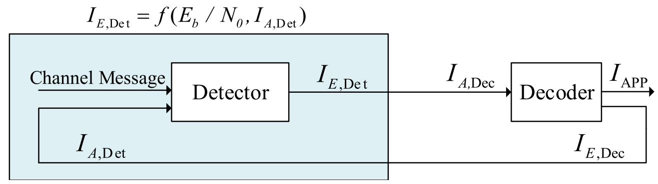

Fig. 4. The MI update for the LDPC-coded FTN system, where the extrinsic MI is exchanged between the detector and the decoder by the outer and inner iteration numbers.

We emphasize an often overlooked but critical point: under identical shaping pulse $h(t)$, the PSDs of both FTN and Nyquist signals are proportional to $|H(f)|^2$. This means that both signals occupy the same bandwidth. While this fact is well-known in theory, it is worth stating explicitly to avoid the common misconception that FTN signaling inherently expands bandwidth due to faster symbol rates.

III. Tailored LDPC Codes Analysis and Design for FTN Signaling

In this section, we present the EXIT chart analysis for the LDPC-coded FTN system. Based on the proposed EXIT chart analysis and simulations, we explore the thresholds and decoding performance of different LDPC codes in both Nyquist and FTN systems, revealing various findings for good LDPC codes in FTN systems. Based on these, we propose tailored LDPC codes for FTN systems.

A. Performance Analysis With EXIT Chart

In this subsection, we present the EXIT chart analysis for the LDPC-coded FTN system. The diagram of EXIT chart analysis is shown in Fig. 4, where $I_{A,\text{Det}}$ and $I_{E,\text{Det}}$ denote the *a priori* MI and the extrinsic MI of the detector, respectively. Similarly, $I_{A,\text{Dec}}$ and $I_{E,\text{Dec}}$ denote the *a priori* MI and the extrinsic MI of the decoder, respectively. To implement the EXIT chart analysis for the LDPC-coded FTN system, we consider the MI update for the LDPC decoder and the FTN detector separately, as described below.

Given the extrinsic MI $I_{E,\text{Det}}$ of the detector (or the *a priori* MI $I_{A,\text{Dec}}$), we can update the MI $I_{E,\text{Dec}}$ over the LDPC code. For conventional regular and irregular LDPC codes, the MI update can be directly calculated for the degree distribution, as described in [27], and the details are omitted here. For the protograph-based LDPC code, we utilize a protograph-based EXIT (PEXIT) chart analysis to characterize the iterative message-passing of MI and check the convergence. Precisely, the base matrix $\boldsymbol{B}$ of a protograph-based LDPC code is given by

$$\boldsymbol{B} = \begin{bmatrix} b_{0,0} & b_{0,1} & \cdots & b_{0,n_b-1} \\ b_{1,0} & b_{1,1} & \cdots & b_{1,n_b-1} \\ \cdots & \cdots & \ddots & \cdots \\ b_{m_b-1,0} & \cdots & \cdots & b_{m_b-1,n_b-1} \end{bmatrix}, \quad (8)$$

where we define the following notations for the PEXIT chart analysis of the protograph-based LDPC codes.

- $I_{Ev}^{i,j}$: the extrinsic MI from the j-th variable node (VN) to the i-th check node (CN).
- $I_{Ec}^{i,j}$: the extrinsic MI from the i-th CN to the j-th VN.
- I_{APP}^{j}: the *a posteriori probability* MI of the j-th VN utilized for convergence check.

The MI update of the PEXIT chart analysis is summarized as follows, where the $J(\cdot)$ and $J^{-1}(\cdot)$ functions are given in [38].

- *Initialization:* For $0 \le i \le m_b - 1, 0 \le j \le n_b - 1$, set the extrinsic MI $I_{Ev}^{i,j} = I_{Ec}^{i,j} = 0$.
- *Extrinsic MI update from the j-th VN to the i-th CN:*

$$I_{Ev}^{i,j} = J\left(\sqrt{\sum_{l\neq i} b_{l,j}\left[J^{-1}(I_{Ec}^{l,j})\right]^2 + [J^{-1}(I_{A,\text{Dec}}^{j})]^2}\right). \tag{9}$$

- *Extrinsic MI update from the i-th CN to the j-th VN:*

$$I_{Ec}^{i,j} = 1 - J\left(\sqrt{\sum_{l\neq j} b_{i,l}\left[J^{-1}(1 - I_{Ev}^{i,l})\right]^2}\right). \tag{10}$$

- *Calculate a posteriori probability (APP) MI and extrinsic MI:* $I_{\text{APP}}^{j} = J\left(\sqrt{[J^{-1}(I_{E,\text{Dec}}^{j})]^2 + [J^{-1}(I_{A,\text{Dec}}^{j})]^2}\right)$, where $I_{E,\text{Dec}}^{j} = J\left(\sqrt{\sum_l b_{l,j}[J^{-1}(I_{Ec}^{l,j})]^2}\right)$.
- *Convergence check:* If $|I_{\text{APP}}^{j} - 1| < \delta, \forall j < n_b - m_b$, where δ is the stopping threshold, a decoding success is declared; Otherwise, declare a decoding failure.

For the FTN detector, there is no direct closed-form or even approximate function between the input $I_{A,\text{Det}}$ and the output $I_{E,\text{Det}}$. A direct approach, as in [39] and [40], is to generate detector extrinsic LLRs via Monte Carlo. While accurate, this requires repeatedly invoking the FTN detector within each PEXIT iteration and across SNR points, which makes threshold computation prohibitively expensive (and virtually impractical) under iterative detection and SNR-point search. In contrast, we approximate the detector MI transfer function via an offline least-squares polynomial fitting from a limited set of simulation points, thereby obtaining the approximation function $I_{E,\text{Det}} \approx f(E_b/N_0, I_{A,\text{Det}})$. Once fitting is complete, the polynomial can be evaluated at negligible cost during threshold computation, thereby eliminating repeated detector calls while still maintaining high accuracy.

Formally, the polynomial fitting function $f(E_b/N_0, I_{A,\text{Det}})$, can be expressed as

$$f(E_b/N_0, I_{A,\text{Det}}) = \sum_{i=0}^{p}\sum_{j=0}^{p} a_{i,j}\left(\frac{E_b}{N_0}\right)^i (I_{A,\text{Det}})^j, \tag{11}$$

where p denotes the highest order of the polynomial. Note that τ is not included in the fitting function, as the fitting is performed for each τ. This fixed-τ fitting maintains analytical simplicity because it approximates a two-dimensional function $f(E_b/N_0, I_{A,\text{Det}})$ for each predefined τ, instead of a more complicated three-dimensional function $f(E_b/N_0, I_{A,\text{Det}}, \tau)$. Moreover, since τ is a predefined system parameter in practical FTN implementations, the required input-output mapping for mutual information can be efficiently obtained using a simple polynomial fit. While the current framework assumes preset τ values, the methodology remains universally applicable, as different τ settings simply require updating the fitting dataset for the corresponding fitting function. Here, the fitting objective is to determine $a_{i,j} (0 \le i \le p, 0 \le j \le p)$ such that $f(E_b/N_0, I_{A,\text{Det}})$ approximates the actual extrinsic MI $I_{E,\text{Det}}$ of the FTN detector (obtained by Monte-Carlo simulations) in the considered E_b/N_0 region and $I_{A,\text{Det}}$ region $(0 \le I_{A,\text{Det}} \le 1)$. This can be achieved by minimizing the sum of squared residuals of polynomial output and $I_{E,\text{Det}}$, which can be formulated as

$$a_{k,l} = \arg\min \sum_{(I_{E,\text{Det}}, I_{A,\text{Det}}, E_b/N_0)\in\Phi} (I_{E,\text{Det}} - f(E_b/N_0, I_{A,\text{Det}}))^2. \tag{12}$$

where $0 \le k \le p, 0 \le l \le p$ and Φ represents the data set from Monte-Carlo simulations.

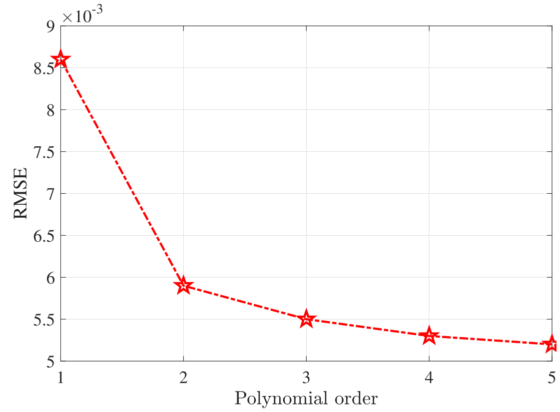

Fig. 5. The RMSE between the polynomial fitting and the actual simulation results for different p.

In this paper, we approximate the mutual information using a two-input polynomial of order p, which has $M = \frac{(p+1)(p+2)}{2}$ coefficients. The fitting cost scales with both p and the number of samples N_s, resulting in a computational complexity of approximately $\mathcal{O}(N_s M^2)$. For example, when $p = 3$ we have $M = 10$; with $N_s = 400$ samples, the offline fitting requires on the order of 4×10^4 operations. Thus, the choice of p reflects an accuracy-complexity tradeoff, which we illustrate with a concrete example below.

Example 1 (RMSE for Different p): Consider an FTN system with $\tau = \frac{2}{3}$, where $I_{E,\text{Det}}$ is approximated using a polynomial of order $p = 3$ based on the large set of samples $I_{A,\text{Det}}$ and E_b/N_0 (from 0 to 3 dB). We evaluate the root mean square error (RMSE) between the polynomial approximation and the actual simulation results for different orders p. The RMSE results are shown in Fig. 1, from which it can be observed that the RMSE essentially converges at $p = 3$. More intuitively, the polynomial fitting function $f(E_b/N_0, I_{A,\text{Det}})$ and the actual results (obtained by Monte-Carlo simulations) with $p = 3$ are illustrated in Fig. 6, from which we can observe that the fitting results can effectively approximate the actual simulation results. In our numerical study, the threshold converges at $p = 3$; increasing p further leads to negligible changes. Accordingly, we set $p = 3$ throughout this paper.

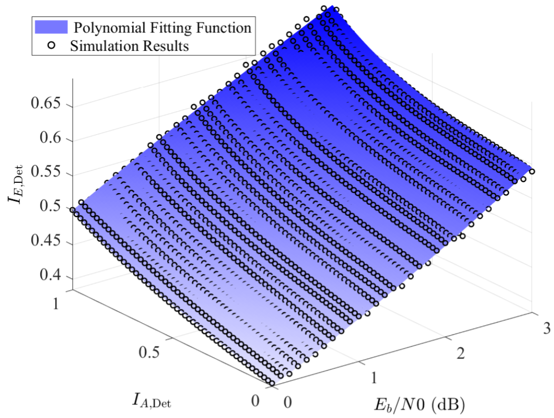

Fig. 6. The polynomial fitting function $f(E_b/N_0, I_{A,\text{Det}})$ and the actual results (sampled by Monte Carlo simulations) with a polynomial of order $p = 3$ and $\tau = \frac{2}{3}$.

Consider the protograph-based LDPC code, which has n_b types of variable nodes. Consequently, the updated extrinsic mutual information $I^j_{E,\text{Dec}}$ $(0 \le j \le n_b - 1)$ varies across different node types. A straightforward approach is to call function $I^j_{E,\text{Dec}} = f(E_b/N_0, I^j_{A,\text{Det}})$ of n_b times for each distinct $I^j_{E,\text{Dec}}$ $(0 \le j \le n_b - 1)$, resulting in n_b distinct detector outputs $I^j_{E,\text{Det}}$ for $0 \le j \le n_b - 1$. However, for simplicity, we adopt an average approximation in this paper, where the MI from the decoder to the detector is given by

$$I_{A,\text{Det}} = \frac{1}{n_b} \sum_{j=0}^{n_b-1} I^j_{E,\text{Dec}}, \tag{13}$$

and hence we only call the function once with $I_{E,\text{Det}} = f(E_b/N_0, I_{A,\text{Det}})$. For LDPC codes with protograph construction, replacing n_b per-type evaluations with one averaged MI is a reasonable approximation for simplification, and similar averaging strategies have also been applied in [40], [41], and [42]. The design of the interleaver, which determines how coded bits are mapped to different modulation bits under higher-order modulation, is also crucial. However, we do not have the space for discussing this important issue in the paper and have to leave this for future work. For the sake of clarity, the proposed PEXIT chart analysis is summarized in Algorithm 1. Furthermore, a bisection method can be performed to determine the decoding threshold $(E_b/N_0)^*$, which is defined as the minimum required E_b/N_0 to achieve decoding success for the LDPC-coded FTN system.

Example 2 (EXIT Chart Curve): Consider the LDPC-coded FTN system with $\tau = \frac{2}{3}$, where the maximum number of outer iterations is $J_{\max} = 25$ and the maximum number of inner iterations is $I_{\max} = 10$. For EXIT chart analysis, we adopt a base graph of size 24×46 from BG1 of 5G. The EXIT curves for the detector and decoder are illustrated in Fig. 2. It can be observed that at $E_b/N_0 = 1$ dB, there is no open tunnel[1] between the detector and decoder, indicating decoding failure. However, at 2 dB, a clear tunnel emerges, indicating successful iterative decoding. These results imply that the decoding threshold, corresponding to the waterfall region, lies within the range of 1 to 2 dB.

[1]By "open tunnel," we refer to the region where the EXIT curves of the decoder and the detector remain separated without intersection, indicating the potential for successful iterative decoding. Further details of open tunnel can be found in [43].

Algorithm 1 PEXIT Chart Analysis for the LDPC-Coded FTN System with Turbo Equalization

Input: Bit signal-to-noise ratio E_b/N_0; Base matrix $\boldsymbol{B}$; Maximum outer iteration number $J_{\max}$; Maximum inner iteration number $I_{\max}$; Stopping threshold δ.
Initialization: Set $I_{E,\text{Det}} = I_{A,\text{Det}} = I^j_{A,\text{Dec}} = I^j_{E,\text{Dec}} = 0$ for $0 \le j \le n_c - 1$.
for $i = 0$ **to** $J_{max} - 1$ **do**
  ○ $I_{A,\text{Det}} \leftarrow \frac{1}{n_b} \sum_{j=0}^{n_b-1} I^j_{E,\text{Dec}}$ $(0 \le j \le n_c - 1)$.
  ○ Take $I_{A,\text{Det}}$ and E_b/N_0 as input and calculate the output extrinsic MI of the detector as $I_{E,\text{Det}} = f(E_b/N_0, I_{A,\text{Det}})$.
  for $l = 0$ **to** $I_{max} - 1$ **do**
    • $I^j_{A,\text{Dec}} \leftarrow I_{E,\text{Det}}$ $(0 \le j \le n_b - 1)$.
    • Take $I^j_{A,\text{Dec}}$ $(0 \le j \le n_b - 1)$ as input and update MI from VNs to CNs according to (9).
    • Update MI from CNs to VNs according to (10).
    • Calculate the a posteriori MI I^j_{APP} and the extrinsic MI of the decoder $I^j_{E,\text{Dec}}$ for $0 \le j \le n_b - 1$.
    if $|I^j_{APP} - 1| < \delta$, $\forall j < n_b - m_b$ **then**
      Report decoding success and return.
    end
  end
end

B. Nyquist Codes and FTN Codes

A critical question in LDPC code design for FTN systems is whether LDPC codes that perform well in Nyquist systems are also suitable for FTN systems. To answer this question, for both Nyquist signaling and FTN signaling, we present the decoding thresholds obtained by the proposed EXIT chart analysis and the decoding performance obtained by simulation for different types of LDPC codes, including regular LDPC codes, irregular LDPC codes, and protograph-based LDPC codes. By comparing the results of rate-1/2 LDPC codes with binary phase shift keying (BPSK)/quadrature phase shift keying (QPSK) constellation and considering the impact of various factors, we derive valuable findings for the design of good LDPC codes for FTN signaling, which further guide the design of tailored LDPC codes for FTN systems. Unless otherwise stated, for FTN systems, the maximum outer iterations is $J_{\max} = 25$, and the maximum inner iterations is $I_{\max} = 10$.

1) Regular LDPC Codes in Nyquist System and FTN System: Consider the common (d_v, d_c)-regular LDPC codes. The decoding thresholds by the proposed EXIT chart and the required E_b/N_0 by simulation (at target BER $= 10^{-4}$ for length $n = 2040$) of the regular LDPC codes are tabulated in Table I. From this table, we observe that regular LDPC codes exhibit similar performance trends in both Nyquist and FTN systems. Specifically, for both Nyquist and FTN signaling, the $(3, 6)$-regular LDPC code performs the best, followed by the

TABLE I
THRESHOLDS (BY EXIT) AND REQUIRED E_b/N_0 (BY SIMULATION WITH BER $= 10^{-4}$) FOR REGULAR LDPC

Codes	Nyquist Signaling (dB)		FTN Signaling (dB)	
	Threshold	Required E_b/N_0	Threshold	Required E_b/N_0
(2,4)-Regular LDPC	3.0375	3.85	3.6920	3.90
(3,6)-Regular LDPC	**1.1017**	**1.96**	**2.1195**	**2.50**
(4,8)-Regular LDPC	1.5340	2.32	2.8446	3.20

TABLE II
THRESHOLDS (BY EXIT) AND REQUIRED E_b/N_0 (BY SIMULATION WITH BER $= 10^{-4}$) FOR IRREGULAR LDPC

Codes	Average d_v	Maximum d_v	Nyquist Signaling (dB)		FTN Signaling (dB)	
			Threshold	Required E_b/N_0	Threshold	Required E_b/N_0
Irregular LDPC-I	3.0514	6	**0.6610**	**1.13**	1.6797	1.72
Irregular LDPC-II	3.0950	5	0.7208	1.19	1.6798	1.80
Irregular LDPC-III	2.8620	4	0.7797	1.22	**1.6690**	**1.70**

TABLE III
ORIGINAL VS. MODIFIED DEGREE DISTRIBUTIONS OF IRREGULAR LDPC CODES

Code	Original Distribution $\Lambda(X)$	Modified Distribution $\Lambda'(X)$
LDPC-I	$0.5072X^2 + 0.2506X^3 + 0.0840X^4 + 0.1582X^6$	$0.5572X^2 + 0.2506X^3 + 0.0840X^4 + 0.1082X^6$
LDPC-II	$0.5053X^2 + 0.1234X^3 + 0.1423X^4 + 0.2290X^5$	$0.6053X^2 + 0.1234X^3 + 0.1423X^4 + 0.1290X^5$
LDPC-III	$0.5488X^2 + 0.0404X^3 + 0.4108X^4$	$0.5988X^2 + 0.0404X^3 + 0.3608X^4$

$(4, 8)$-regular LDPC code, and the $(2, 4)$-regular LDPC code shows the worst performance.

2) Irregular LDPC Codes in Nyquist System and FTN System: Consider three types of irregular LDPC codes from [27], with the following variable-node degree distribution $\Lambda(x)$ as: 1) irregular LDPC-I with $\Lambda(x) = 0.5072X^2 + 0.2506X^3 + 0.0840X^4 + 0.1582X^6$;2) irregular LDPC-II with $\Lambda(x) = 0.5053X^2 + 0.1234X^3 + 0.1423X^4 + 0.2290X^5$;3) irregular LDPC-III with$\Lambda(x) = 0.5488X^2 + 0.0404X^3 + 0.4108X^4$. The decoding thresholds by the proposed EXIT chart and the required E_b/N_0 by simulation (at target BER $= 10^{-4}$ for length $n = 15360$) are tabulated in Table II, from which we observe that irregular LDPC-I achieves the best performance and the lowest decoding threshold in the Nyquist system, while in the FTN system, irregular LDPC-III performs best. We also provide the average degree d_v in the table, where it can be observed that the decoding performance in the FTN system is closely related to the average degree. Specifically, the irregular LDPC-III, which has the smallest average degree, achieves the best performance, while the irregular LDPC-II, with the largest average degree, exhibits the worst performance. This observation suggests that LDPC codes optimized for FTN systems are generally sparser than those designed for Nyquist systems. This is because the presence of ISI inherently introduces dependencies between coded bits, and a relatively dense parity-check matrix design can reduce the accuracy of LLR messages in BP decoding. Nevertheless, excessive sparsity may also lead to degraded performance.

3) Modified Irregular LDPC Codes for Sparsity: To further reinforce the insight that appropriate sparsity is critical in FTN systems, we have extended by conducting a degree-distribution optimization of the three irregular LDPC codes. In this optimization, our goal is to reduce the average variable-node degree. One solution is to decrease the proportion of high-degree variable nodes and correspondingly increase that of low-degree ones. However, since finding the globally optimal degree distribution constitutes a complex combinatorial search problem, we simplify the process by adjusting only the relative proportions of the largest and smallest degree nodes. Formally, let the original variable-node degree distribution be $\Lambda(X) = \sum_d \Lambda_d X^d$. Denote the maximum variable degree by $d_{\max}$ (with weight $\Lambda_{d_{\max}}$) and the minimum variable degree by $d_{\min}$ (with weight $\Lambda_{d_{\min}}$). The optimization proceeds iteratively as follows: at each step, update as $\Lambda_{d_{\max}} \leftarrow \Lambda_{d_{\max}} - \Delta, \Lambda_{d_{\min}} \leftarrow \Lambda_{d_{\min}} + \Delta$, with a fixed increment $\Delta = 0.05$, re-normalize if necessary to ensure $\sum_d \Lambda_d = 1$, and compute the corresponding EXIT threshold. We retain the best candidate (smallest threshold) encountered so far. The sweep stops when $\Lambda_{d_{\max}} \leq 0$ (or falls below numerical tolerance), and the distribution achieving the minimum threshold over the sweep is selected. The original and optimized variable-node distributions are summarized in Table III.The decoding thresholds by the proposed EXIT chart and the required E_b/N_0 by simulation (at target BER $= 10^{-4}$ for length $n = 15360$) for these three irregular LDPC codes are tabulated in Table IV, V and VI, from which we observe that the original

TABLE IV

THRESHOLDS (BY EXIT) AND REQUIRED E_b/N_0 (BY SIMULATION WITH BER $= 10^{-4}$) FOR (MODIFIED) IRREGULAR LDPC-I

Codes	Average d_v	Maximum d_v	Nyquist Signaling (dB)		FTN Signaling (dB)	
			Threshold	Required E_b/N_0	Threshold	Required E_b/N_0
Irregular LDPC-I	3.0514	6	**0.6610**	**1.13**	1.6797	1.72
Modified irregular LDPC-I	2.8514	6	0.7667	1.22	**1.5508**	**1.62**

TABLE V

THRESHOLDS (BY EXIT) AND REQUIRED E_b/N_0 (BY SIMULATION WITH BER $= 10^{-4}$) FOR (MODIFIED) IRREGULAR LDPC-II

Codes	Average d_v	Maximum d_v	Nyquist Signaling (dB)		FTN Signaling (dB)	
			Threshold	Required E_b/N_0	Threshold	Required E_b/N_0
Irregular LDPC-II	3.0950	5	**0.7092**	**1.19**	1.6798	1.80
Modified irregular LDPC-II	2.7950	5	1.1121	1.22	**1.5785**	**1.62**

TABLE VI

THRESHOLDS (BY EXIT) AND REQUIRED E_b/N_0 (BY SIMULATION WITH BER $= 10^{-4}$) FOR (MODIFIED) IRREGULAR LDPC-III

Codes	Average d_v	Maximum d_v	Nyquist Signaling (dB)		FTN Signaling (dB)	
			Threshold	Required E_b/N_0	Threshold	Required E_b/N_0
Irregular LDPC-III	2.8620	4	**0.7797**	**1.22**	1.6690	1.70
Modified irregular LDPC-III	2.7620	4	0.9784	1.30	**1.6384**	**1.65**

TABLE VII

THRESHOLDS (BY EXIT) AND REQUIRED E_b/N_0 (BY SIMULATION WITH BER $= 10^{-4}$) FOR 5G LDPC WITH INFORMATION PUNCTURE

Codes	Nyquist Signaling (dB)		FTN Signaling (dB)	
	Threshold	Required E_b/N_0	Threshold	Required E_b/N_0
standard 5G LDPC	**0.4296**	**0.84**	1.5625	1.93
5G LDPC without information puncture	0.6637	1.05	**1.3413**	**1.74**

three LDPC codes exhibits better performance for Nyquist signaling,whereas the modified LDPC codes (featuring sparser Tanner graphs) demonstrates enhanced performance for FTN signaling. This result suggests that, within a given structure, LDPC codes with an appropriate level of sparsity tend to perform better in FTN systems.

Remark: Average degree alone does not determine performance in coded FTN systems. The degree distribution—particularly the fraction of degree-2 variable nodes—and the structure of the Tanner graph also play a significant role in determining both the waterfall threshold and error floor. Our irregular design performs better than the $(2,4)$-regular code because it markedly reduces the degree-2 fraction, which leads to lower thresholds. In this paper, "sparser parity-check matrices" is intended as a design tendency within comparable ensembles or given LDPC codes, rather than an absolute rule across different families. Under a fixed degree-distribution shape, modest reductions in the average degree can achieve additional FTN gains (see Table IV–VI)—too sparse can also hurt performance.

4) 5G NR Information Puncturing in Nyquist System and FTN System: Consider the 5G LDPC code with and without information puncturing in both FTN and Nyquist systems. The decoding thresholds by the proposed EXIT chart and the required E_b/N_0 by simulation (at target BER $= 10^{-4}$ for length $n = 15360$) are tabulated in Table VII. From this table, we observe that the 5G NR information puncturing, which is commonly used to enhance performance in Nyquist signaling, leads to significant performance loss in the FTN system. This is not surprising, under FTN, ISI induces correlations that reduce the reliability of the detector's extrinsic messages compared with Nyquist. Punctured bits further suffer from the absence of direct channel LLRs and their denser connection of the 5G protograph, so their a posteriori MI increases more slowly, hindering recovery and degenerating performance. Note that the punctured and unpunctured LDPC codes compared in Table VII have the same code rate and code length, ensured by puncturing and shortening. The standard 5G design employs a base graph of size $(m_b+2)\times(n_b+2)$ with information puncturing. By contrast, our unpunctured construction

TABLE VIII

THRESHOLDS (BY EXIT) AND REQUIRED E_b/N_0 (BY SIMULATION WITH BER $= 10^{-4}$) FOR DIFFERENT GIRTHS

Codes	Girth g	Nyquist Signaling (dB)		FTN Signaling (dB)	
		Threshold	Required E_b/N_0	Threshold	Required E_b/N_0
(3,6)-Regular LDPC	4	1.1017	2.24	2.1154	2.70
(3,6)-Regular LDPC	6		1.96		**2.50**
(3,6)-Regular LDPC	8		**1.94**		2.51

is based on a reduced protograph of size $m_b \times n_b$. After applying the same lifting factor Z, the resulting unpunctured LDPC code attains the same design rate as the standard 5G LDPC codes, while preserving the standard diagonal structure. Specifically, the punctured 5G LDPC code is generated from BG1 of size 24×46 with the first two columns punctured, whereas the unpunctured code is generated from BG1 of size 22×44.

5) Impact of Girth in Nyquist System and FTN System: Consider the $(3,6)$-regular LDPC codes with different girths of 4, 6, and 8, where the LDPC codes with girth 6 and girth 8 were both constructed using the PEG algorithm. The decoding thresholds by the proposed EXIT chart and the required E_b/N_0 by simulation (at target BER $= 10^{-4}$ for length $n = 2040$) are tabulated in Table VIII. It can be observed that, similar to Nyquist signaling, LDPC code design for FTN systems also requires avoiding short cycles (with girth $g \geq 6$).

The above results indicate that LDPC codes that perform well in Nyquist systems are not necessarily optimal for FTN systems. There are three observational findings from the above results for designing effective LDPC codes for FTN systems, as summarized below.

- For the tested regular LDPC codes, the relative error-rate ranking under FTN signaling remains consistent with that under Nyquist signaling.
- Different from Nyquist signaling, LDPC codes tailored for FTN signaling often exhibit relatively sparser parity-check matrices, as observed in tested examples. Specifically, within comparable ensembles or for a given protograph shape, modest reductions in the average degree can achieve additional gain for FTN. Nevertheless, sparsity is only one aspect of code design, and an overly sparse structure can also degrade performance. There exists an inherent tradeoff in the sparsity of the parity-check matrix for FTN signaling: reducing the density can improve the girth, whereas an excessively sparse matrix may degrade the decoding performance. In our design, a good tradeoff is achieved by the proposed EXIT chart analysis for the LDPC-coded FTN system. Concretely, for the base graph of the 5G LDPC, we perform multiple random masking trials, evaluate the decoding threshold of each candidate via EXIT analysis, and then select the masking pattern that yields the smallest threshold. That is, the appropriate sparsity level is implicitly set by EXIT-guided masking via selection of the masking pattern with the lowest decoding threshold.
- Different from Nyquist signaling, the 5G NR information puncturing for 5G LDPC codes, which is commonly employed in Nyquist signaling to enhance performance, is detrimental in FTN signaling.

Remark: In all the tables, it is observed that both the decoding threshold and the required SNR for the FTN system are worse than those for the Nyquist system. This is because the same LDPC code is used for both FTN signaling and Nyquist signaling to investigate performance trends. Under this setup, the FTN system inherently achieves higher SE, which leads to higher thresholds and required SNRs. For a fair comparison in practical scenarios, FTN and Nyquist systems should operate under the same SE. In such cases, the FTN system allows for the use of lower-rate LDPC codes while maintaining the same overall bandwidth. This highlights a key advantage of LDPC-coded FTN systems—enabling the deployment of lower-rate codes, which can significantly improve reliability.

C. Tailored LDPC Code Design for FTN System

Motivated by the aforementioned findings for designing good LDPC codes for FTN systems, we consider the tailored design of LDPC codes for FTN signaling. Specifically, for practical consideration, we consider three cases for comparison, as outlined below:

- $\mathcal{C}_1$: The standard 5G LDPC code, where the first $2Z$ (corresponding to the first two columns of the base graph) information bits are punctured with a base graph size of $(m_b + 2) \times (n_b + 2)$.
- $\mathcal{C}_2$: The same base graph structure as the standard 5G LDPC code but with a reduced base graph of size $m_b \times n_b$, where the first $2Z$ bits are not punctured, resulting in the same rate as 5G standard.
- $\mathcal{C}_3$: Based on $\mathcal{C}_2$ (with size of $m_b \times n_b$), we apply a random masking operation to the base graph, where the masking process is guided by the proposed EXIT chart analysis, ensuring that the resulting tailored LDPC code achieves a lower decoding threshold. Subsequently, the base matrix is lifted according to the 5G standard, generating the parity-check matrix for the improved 5G LDPC codes.

The procedure of the masking-based codes $\mathcal{C}_3$ is summarized as follows: i) Set the maximum number of search iterations $\ell_{\max}$. ii) In each iteration, draw a masking probability $p \sim U[p_{\min}, p_{\max}]$. Perform independent Bernoulli-p trials for each non-negative entry in the $\mathbf{A}$ and $\mathbf{E}$ parts of the 5G base matrix to decide whether it should be masked,

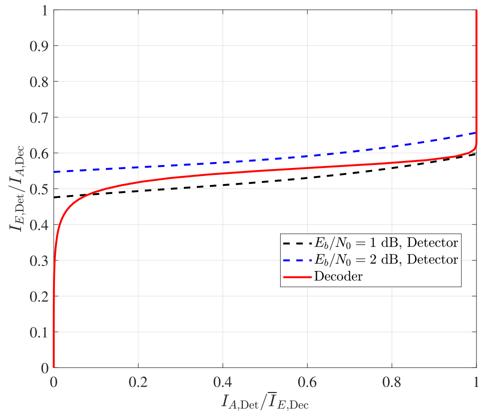

Fig. 7. The EXIT curves of the detector and decoder for the LDPC-coded FTN system, where a base graph of size 24×46 from BG1 of 5G is adopted.

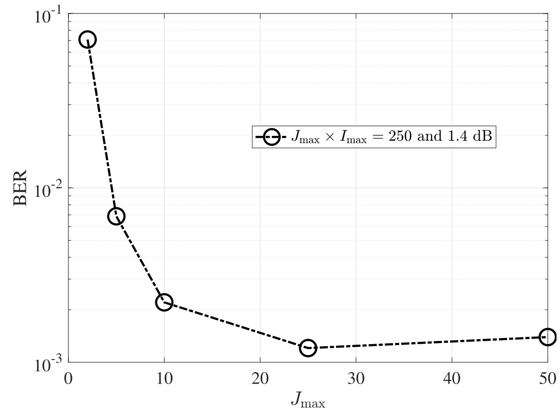

Fig. 9. The simulation results for different inner and outer iterations with $J_{\max} \times I_{\max} = 250$ for 1.4 dB.

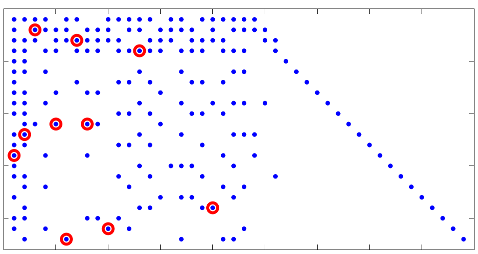

Fig. 8. The masked BG1 with size 22×44, where all blue points form the scatter diagram of the standard 5G LDPC code, while the red-circled points indicate the locations where masking is applied.

while preserving the diagonal structure $\mathbf{I}$ and the double-diagonal part $\mathbf{D}$ to ensure compatibility with the 5G standard. iii) Compute the decoding threshold of the masked base matrix, and retain the matrix that yields the smallest threshold observed so far. iv) Repeat steps ii)-iii) until $\ell_{\max}$ iterations are reached, and output the retained masked base matrix. To provide a clearer illustration of the masking operation, we present the masked and optimized base graph of size 22×44 derived from BG1, obtained with $\ell_{\max} = 500$, $p_{\min} = 0.01$, and $p_{\max} = 0.1$. The optimized base graph is shown in Fig. 8, where all blue points form the scatter diagram of the standard 5G LDPC code, while the red-circled points indicate the locations where masking is applied and result in zero matrices for these points after lifting.

Remark: For the proposed LDPC codes $\mathcal{C}_3$, the masking operation reduces the density of the parity-check matrix, effectively decreasing the number of edges in its Tanner graph. Consequently, the Tanner graph of the newly obtained code contains fewer short cycles or exhibits a larger girth [44]. As a result, the proposed LDPC code $\mathcal{C}_3$ naturally inherits this short cycle-free property of the standard 5G for long codes and even has a large girth. We also note that in the threshold-based masking search, the random masking is applied only to the $\mathbf{A}$ and $\mathbf{E}$ parts of the 5G base matrix, while the single-diagonal $\mathbf{I}$ and double-diagonal $\mathbf{D}$ parts remain unchanged. This ensures that the diagonal structure is preserved and compatibility with the 5G standard is maintained. A fixed number of trials is performed, and the protograph with the smallest EXIT threshold among the candidates is selected. To ensure 5G compatibility, the proposed FTN-oriented codes are obtained by applying masking to the original 5G base graphs, while preserving the raptor-like $\mathbf{I}/\mathbf{D}$ diagonal structure and the standardized lifting factors. Consequently, the design freedom is intentionally constrained by implementation considerations, making the constructed codes inherently suboptimal by construction; nevertheless, they provide a good trade-off between performance and 5G compatibility.

In FTN systems, the equalizer/detector information exchange plays an important role due to the strong ISI [45]. In practice, using fewer inner iterations avoids overly confident decoder feedback and improves global convergence, which aligns with observations in turbo equalization systems. Hence, we prioritize outer iterations over inner LDPC iterations. Concretely, we adopt $J_{\max} = 25$ outer iterations and $I_{\max} = 10$ inner iterations. This setting is selected by fixing the total iteration budget $J_{\max} I_{\max} = 250$ and sweeping $(J_{\max}, I_{\max})$. The BER performance is shown in Fig. 9 for the LDPC-coded FTN with BPSK ($\tau = \frac{2}{3}$, proposed LDPC code with $n = 15360$ and $r = \frac{1}{2}$) at $E_b/N_0 = 1.4$ dB. It can be observed that the best is achieved with $J_{\max} = 25$ and $I_{\max} = 10$. Accordingly, this configuration is adopted in this paper.

Example 3 (EXIT Curves for $\mathcal{C}_1$, $\mathcal{C}_2$ and $\mathcal{C}_3$): Consider the EXIT chart analysis of the LDPC-coded FTN system with $\tau = \frac{2}{3}$, where $\mathcal{C}_1$, $\mathcal{C}_2$, and $\mathcal{C}_3$ are adopted here. The EXIT curves of decoder and the detector at $E_b/N_0 = 1.2$ dB are illustrated in Fig. 10. It can be observed that at 1.2 dB, there is no open tunnel between the EXIT curves of the decoder (for $\mathcal{C}_1$ and $\mathcal{C}_2$) and the detector, indicating that these codes do not converge at 1.2 dB. However, for $\mathcal{C}_3$, an open tunnel is observed, indicating that at this SNR, a decoding strategy exists, allowing iterative BP decoding can be successful. This analysis implies that the proposed code $\mathcal{C}_3$ achieves a lower threshold compared to standard 5G LDPC codes.

Example 4 (Thresholds and Decoding Performance of $\mathcal{C}_1, \mathcal{C}_2$ and $\mathcal{C}_3$): Consider the LDPC-coded FTN system with $\tau = \frac{2}{3}$, where the maximum number of outer iterations is $J_{\max} = 25$ and the maximum number of inner iterations is $I_{\max} = 10$. $\mathcal{C}_1, \mathcal{C}_2$ and $\mathcal{C}_3$ are adopted with the same $k = 7680$

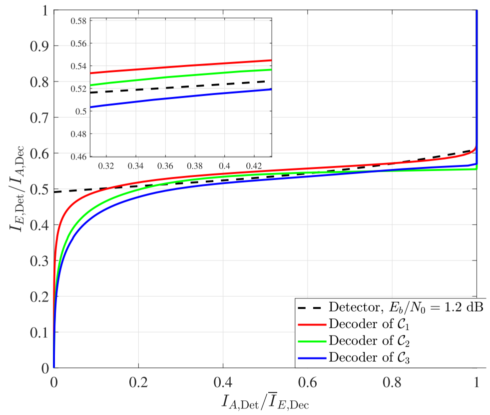

Fig. 10. The EXIT curves of decoder and the detector at $E_b/N_0 = 1.2$ dB.

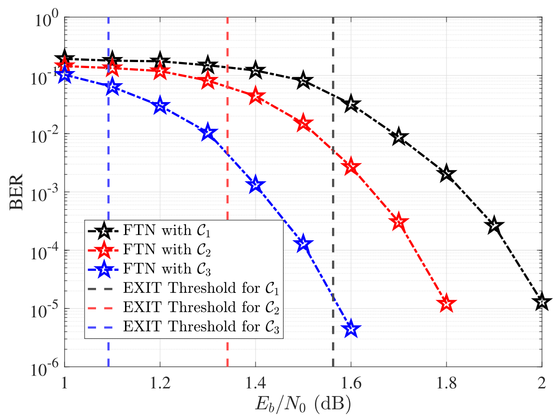

Fig. 11. The decoding performance by simulation and the decoding thresholds by PEXIT chart analysis of the coded FTN system with $\tau = \frac{2}{3}$, where $\mathcal{C}_1$ (standard 5G LDPC), $\mathcal{C}_2$ (5G LDPC without information puncture) and $\mathcal{C}_3$ (tailored LDPC) are adopted, all with $k = 7680$ and $r = 0.5$. The corresponding decoding thresholds are approximately 1.5625 dB for $\mathcal{C}_1$, 1.3413 dB for $\mathcal{C}_2$, and 1.0919 dB for $\mathcal{C}_3$.

and $r = 0.5$, resulting in the SE of 0.75 bits per dimension. The decoding thresholds obtained by the proposed EXIT chart analysis, along with the corresponding simulation results, are shown in Fig. 11. As expected, $\mathcal{C}_3$ (the proposed LDPC code) achieves the lowest threshold of about 1.0919 dB and thus the best performance, followed by $\mathcal{C}_2$ with a threshold of about 1.3413 dB, while $\mathcal{C}_1$ exhibits the highest threshold of about 1.5625 dB and performs worst. Specifically, a coding gain over 0.4 dB at BER $= 10^{-5}$ is achieved by the proposed LDPC codes compared to the standard 5G LDPC code for the FTN system. For all three codes, we observe that the simulation curves and EXIT thresholds intersect near $\mathrm{BER} \approx 5 \times 10^{-2}$. This consistency is expected: the EXIT threshold characterizes the waterfall region in the infinite-blocklength regime, whereas our simulations are at finite blocklength.

Remark: A lower decoder EXIT curve for $\mathcal{C}_3$ under FTN does not, in isolation, imply superior performance in Nyquist. Under FTN, the decoder curve is obtained with ID and is influenced by the detector; in Nyquist (notably with BPSK/QPSK), the decoder-only PEXIT curve (PEXIT without detector-decoder iteration) is the appropriate reference as detector-decoder iteration yields no gain. Consistent with this distinction, our decoding thresholds for Nyquist

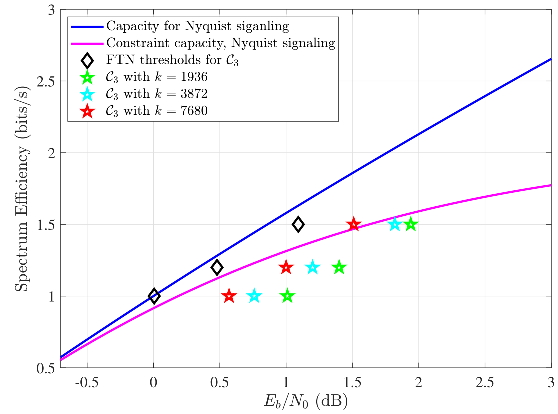

Fig. 12. The asymptotic thresholds and finite-length performance with different spectral efficiencies, where the information bit lengths of $k = 1936$, 3872, and 7680, and code rates of $r = 1/3$, 2/5 and 1/2 are adopted.

system under AWGN channels show $\mathcal{C}_1$ performs best (about 0.4296 dB), followed by $\mathcal{C}_2$ (about 0.6637 dB), with $\mathcal{C}_3$ the worst (about 0.8359 dB)—an ordering different from that under FTN—reflecting the intended FTN-Nyquist tradeoff.

Example 5 (Asymptotic Thresholds and Finite-Length Performance): Consider the LDPC-coded FTN system with $\tau = 2/3$ and QPSK modulation. The decoding threshold by the proposed EXIT chart and the required E_b/N_0 (with a target BER $= 10^{-4}$) by simulation is shown in Fig. 12. We present the required E_b/N_0 at a target BER $= 10^{-4}$ for different SE and the set-ups are given as follows: i) SE of 1 bits/s by $r = 1/3$; ii) SE of 1.2 bits/s by $r = 2/5$; iii) SE of 1.5 bits/s by $r = 1/2$. For each SE configuration, we utilize the designed LDPC codes $\mathcal{C}_3$ with $k = 1936$ (lifting factor $Z = 88$), 3872 (lifting factor $Z = 176$), and 7680 (lifting factor $Z = 352$). For comparison, we also plot the constrained capacity for QPSK of Nyquist signaling, which represents the capacity of a Nyquist system with a finite input alphabet. From this figure, we can observe that as the code length increases, the error performance converges toward the asymptotic thresholds of the codes, though a gap remains. We also observe that the asymptotic threshold can closely approach the Shannon capacity of Nyquist signaling, and the decoding performance with $k = 7680$ can approach (or slightly exceed) the constrained capacity of a Nyquist system.

IV. Numerical Result

In this section, we present the numerical results of the proposed LDPC code with FTN signaling under various parameter configurations, which are compared with the Nyquist counterpart operating at equivalent SE. Unless otherwise stated, for FTN systems, the maximum outer iterations are $J_{\max} = 25$, and the maximum inner iterations are $I_{\max} = 10$.

Example 6 (Tailored LDPC-Coded FTN Scheme for Different τ): Consider the LDPC-coded FTN system with BPSK modulation, where the FTN compression factor is $\tau = 2/3$ and 1/2. The tailored LDPC codes $\mathcal{C}_3$ with length of $n = 10560$ are adopted for FTN signaling, where the code rate is $r = 1/2$ (lifting factor $Z = 240$) for $\tau = 2/3$ and $r = 2/5$ (lifting factor $Z = 192$) for $\tau = 1/2$ to achieve different SE. We need to point out that the LDPC codes constructed for

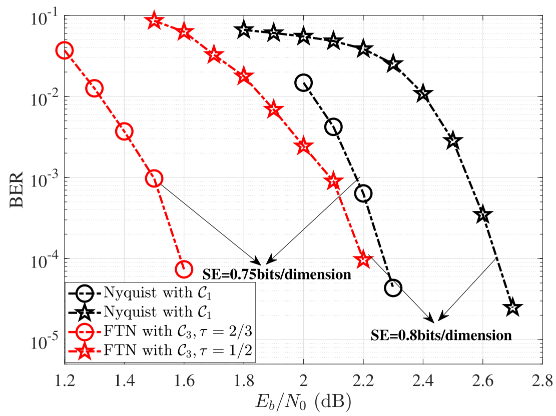

Fig. 13. The performance comparison between the tailored LDPC-coded FTN (for different τ) and the 5G LDPC-coded Nyquist under the same SE constraint, where $\mathcal{C}_1$ of $n = 10560$ is the standard 5G LDPC code and $\mathcal{C}_3$ of $n = 10560$ is the proposed LDPC.

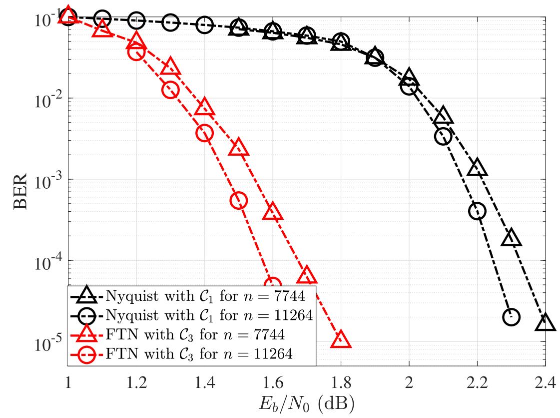

Fig. 14. The performance comparison between the tailored LDPC-coded FTN (for different n) and the 5G LDPC-coded Nyquist under the same SE, where $\mathcal{C}_1$ of $n = 7744$ and 11264 are the standard 5G LDPC codes and $\mathcal{C}_3$ of $n = 7744$ and 11264 are the proposed LDPC codes for FTN signaling.

different FTN compression factors τ are distinct, as different τ values introduce different levels of ISI. Nevertheless, the overall design framework and the protograph-based construction method remain the same across different τ values. For Nyquist signaling, the standard 5G LDPC codes with the same length $n = 10560$ and code rates of $r = 3/4$ (lifting factor $Z = 384$) and $r = 4/5$ (lifting factor $Z = 384$) are selected to achieve the same SE as the corresponding FTN scheme. The simulation results are presented in Fig. 13, from which we observe that, for different SE, FTN signaling with the proposed LDPC codes outperforms Nyquist signaling with the standard 5G LDPC codes under the same SE constraint, achieving a coding gain over 0.6 and 0.4 dB at BER $= 10^{-4}$ for 0.75 and 0.80 bits per dimension, respectively. The results show that the tailored LDPC codes with FTN signaling are universal for different compression factors τ.

Example 7 (Tailored LDPC-Coded FTN Scheme for Different Length n): Consider the LDPC-coded FTN system with BPSK modulation and the compression factor of $\tau = 2/3$. For FTN signaling, the proposed LDPC codes with lengths of $n = 7744$ (lifting factor $Z = 176$) and 11264 (lifting factor $Z = 256$) are adopted, both with a code rate of $r = 1/2$. To achieve the same SE as the FTN scheme, the standard 5G LDPC codes with identical lengths $n = 7744$ (lifting factor $Z = 288$) and 11264 (lifting factor $Z = 384$), but both with a code rate of $r = 3/4$, are adopted for the Nyquist signaling. The simulation results are shown in Fig. 14, from which we observed that for different lengths n, FTN signaling with the proposed LDPC codes $\mathcal{C}_3$ outperforms Nyquist signaling with the standard 5G LDPC codes under the same SE constraint, achieving a coding gain of up to 0.6 dB at BER $= 10^{-4}$.

Example 8 (Tailored LDPC-Coded FTN Scheme for 16-QAM): Consider the LDPC-coded FTN system with 16-quadrature amplitude modulation (QAM) constellation and compression factor of $\tau = 2/3$. The proposed LDPC codes $\mathcal{C}_3$ with lengths of $n = 7744, 9152$ and 11264 are adopted, both with a code rate of $r = 1/2$ for FTN signaling. For Nyquist signaling, we employ the standard 5G LDPC codes with the same $n = 7744$ (lifting factor $Z = 288$), 9152 (lifting factor $Z = 208$) and 11264 (lifting factor $Z = 384$), both

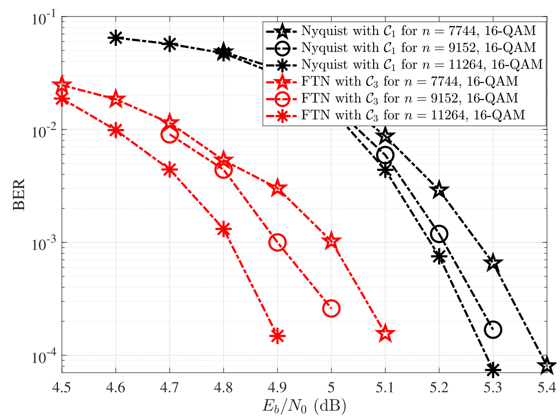

Fig. 15. The performance of the FTN system and Nyquist signaling at 3 bits per channel use with 16-QAM, where the tailored LDPC codes with a rate of 1/2 and $n = 7744, 9152$ and 11264 are employed for the FTN system, while the original LDPC codes with a rate of 3/4 and $n = 7744, 9152$ and 11264 are utilized for Nyquist signaling.

with a code rate of $r = 3/4$, to achieve the same SE with the FTN system. The BICM-ID scheme is also employed for the Nyquist system, where the numbers of outer and inner iterations are set identically to those in the FTN system. The results are illustrated in Fig. 15, from which we observe that, for 16-QAM modulation, FTN signaling with the proposed codes also achieves a coding gain about over 0.3 dB at BER $= 10^{-3}$ compared to Nyquist signaling with the standard 5G LDPC, at the same SE.

Example 9 (Finite-Length Performance and Capacity): Consider the LDPC-coded FTN system with $\tau = 2/3$ and the LDPC-coded Nyquist system, where the information length is $k = 5632$ (lifting factor $Z = 256$) and the modulation is QPSK. We present the required E_b/N_0 at a target BER $= 10^{-4}$ for different SE and the set-ups are given as follows: i) SE of 1 bits/s: FTN system with $\mathcal{C}_3$ of $r = 1/3$ and Nyquist system with $\mathcal{C}_1$ of $r = 1/2$; ii) SE of 1.2 bits/s: FTN system with $\mathcal{C}_3$ of $r = 2/5$ and Nyquist system with $\mathcal{C}_1$ of $r = 3/5$; iii) SE of 1.5 bits/s: FTN system with $\mathcal{C}_3$ of $r = 1/2$ and Nyquist system with $\mathcal{C}_1$ of $r = 3/4$. For the same SE, the code length is consistent between the Nyquist and FTN systems. We also present the capacity under complex Gaussian inputs with

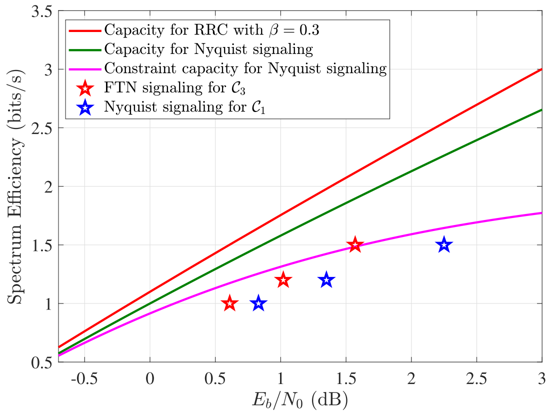

Fig. 16. The capacity and the finite-length performance with different spectral efficiencies for the Nyquist and FTN systems, where $\mathcal{C}_1$ (standard 5G LDPC) and $\mathcal{C}_3$ (tailored LDPC) with $k = 5632$ are adopted.

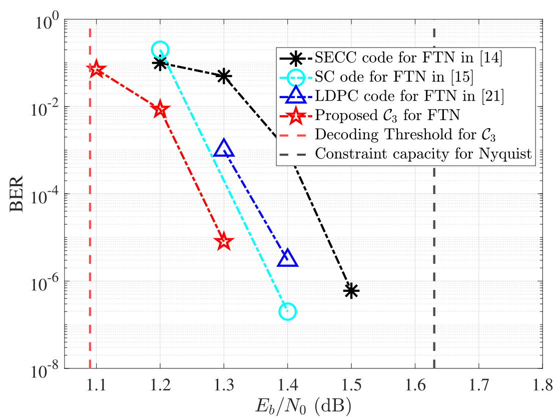

Fig. 17. The performance comparison between the proposed LDPC codes and the existing high-performance codes, including the LDPC codes in [22], the self-concatenated convolutional code in [15], and the spatially coupled FTN code in [16], for the same FTN signaling. All codes have a comparable length with $n \approx 64800$ and code rate 1/2 for a fair comparison.

a roll-off factor of $\beta = 0.3$, along with the capacity for Nyquist under complex Gaussian inputs. The results are illustrated in Fig. 16, from which we observe that FTN signaling with the proposed LDPC codes outperforms the standard 5G LDPC-coded Nyquist system under the same SE constraint. We also observe that the decoding performance at 1.5 bits/s can approach (or slightly exceed) the constrained Nyquist capacity for QPSK, while at lower spectral efficiencies, a gap remains. This observation aligns with the general trend of capacity gain.

Example 10 (Performance Comparison With Existing High-Performance Codes Design): Consider the LDPC-coded FTN system with $\tau = 2/3$, where a rate-1/2 proposed LDPC code $\mathcal{C}_3$ with $n = 64680$ is adopted. Here, we need to point out that while the maximum k in standard 5G is 8448, we proportionally scale the 5G lifting factor ($Z = 1470$) to achieve $n = 64680$ for the proposed LDPC codes to fairly compare with other schemes under similar n and code rate. For comparison, we also plot the performance of the LDPC code for the FTN system as [22], the self-concatenated convolutional code as [15] and the spatially coupled FTN code as [16]. All codes have a comparable length of $n \approx 64680$ and the same code rate of 1/2 for a fair comparison. The results are illustrated in Fig. 17, from which we observe that the proposed LDPC code outperforms existing high-performance schemes for the same FTN signaling.

V. Conclusion

In this paper, by approximating the input-output mutual information function of the FTN detector, we have proposed an EXIT chart analysis for the LDPC-coded FTN system. Based on the proposed EXIT chart analysis, we present the decoding thresholds (along with simulation results) for different LDPC codes in the FTN system compared to the Nyquist system. These results suggest that good LDPC codes for FTN should avoid 5G NR-like information puncturing and operate at an appropriate sparsity level; based on our experiments, this level is generally somewhat sparser than that of Nyquist-oriented designs because of the inherent ISI. Furthermore, we proposed tailored improved 5G LDPC codes for the FTN signaling according to the findings that achieve a lower decoding threshold and better decoding performance compared to the standard 5G LDPC codes, while maintaining the raptor-like structural features and also supporting a reusable encoder/decoder. Our numerical results demonstrate significant coding gains by the proposed improved 5G LDPC codes over the standard 5G LDPC codes in the FTN system. Moreover, the proposed improved 5G LDPC codes with FTN signaling also show better performance than the well-designed 5G LDPC with Nyquist signaling. In addition, the proposed improved 5G LDPC codes achieve better performance compared to the existing high-performance schemes designed for FTN signaling. This validates the effectiveness and potential of the proposed LDPC codes for energy-efficient FTN transmissions in future 6G communications. While this work adopts random masking, exploring other methods such as genetic algorithms to reduce offline search complexity is a future work. Moreover, a more precise analysis would need to take into account the lifted code, including the edge permutations, instead of the protograph alone. The joint protograph-lifting optimization is an interesting and important topic for future work.

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